Hints: Trigonometry – 2. Roots of unity and the complex exponential

Homework

The hints to each problem are cumulative. Read one, try again, and open the next only if you are still stuck.

Problem 1

Hint 1. Put $\zeta = e^{2\pi i/p}$ and write $|G|^2 = G\bar{G} = \sum_{j,k} \zeta^{j^2-k^2} = \sum_{j,k} \zeta^{(j-k)(j+k)}$.

Hint 2. Since p is odd, $(j,k) \mapsto (u,v) = (j-k, j+k)$ is a bijection of $(\mathbb{Z}/p)^2$. So $|G|^2 = \sum_u \sum_v \zeta^{uv}$, and by the root-of-unity filter the inner sum is p for $u = 0$ and 0 otherwise.

Hint 3. Each nonzero square is hit by two values of k, so $G = \sum_{a=0}^{p-1}(1 + (\frac{a}{p}))\zeta^a = \sum_a (\frac{a}{p})\zeta^a$. Substituting $a \to -a$ gives $\bar{G} = (\frac{-1}{p})G$, hence $G^2 = (\frac{-1}{p})|G|^2 = (\frac{-1}{p})\, p = (-1)^{(p-1)/2}p$.

Problem 2

Hint 1. The numbers $1, \cos\frac{2\pi}{7}, \cos\frac{4\pi}{7}, \cos\frac{6\pi}{7}$ span the field $K = \mathbb{Q}(\cos\frac{2\pi}{7})$, which has degree 3 over $\mathbb{Q}$. So the question is whether $\sqrt[3]{2} \in K$, resp. $\sqrt[5]{2} \in K$.

Hint 2. For $\sqrt[5]{2}$: its degree is 5 (Eisenstein), which does not divide 3. Answer: no.

Hint 3. For $\sqrt[3]{2}$ degrees do not help. Use symmetry instead: $c_j = 2\cos\frac{2\pi j}{7}$ $(j = 1, 2, 3)$ are the roots of $x^3 + x^2 - 2x - 1$ and $c_2 = c_1^2 - 2$, so $c_1 \mapsto c_2 \mapsto c_3 \mapsto c_1$ is a field automorphism σ of K (every element of K is a polynomial in c_1). The fixed elements of σ are exactly the rationals.

Hint 4. If $\sqrt[3]{2} \in K$, then $\sigma(\sqrt[3]{2})$ is a root of $x^3 - 2$ lying in the real field K, hence equal to $\sqrt[3]{2}$; so $\sqrt[3]{2}$ would be rational. Answer: no. The two answers come from different obstructions: the degree for $\sqrt[5]{2}$, and for $\sqrt[3]{2}$ the fact that K is a normal (cyclic) extension while $\mathbb{Q}(\sqrt[3]{2})$ is not.

Problem 3

Hint 1. From $\cos(n+1)\theta + \cos(n-1)\theta = 2\cos\theta\cos n\theta$ get $T_{n+1} = 2xT_n - T_{n-1}$, $T_0 = 1$, $T_1 = x$; induction gives integer coefficients and leading coefficient 2^{n-1} (for $n \geqslant 1$).

Hint 2. $D_n(x) = 2T_n(x/2)$ is monic with integer coefficients and $D_n(2\cos\theta) = 2\cos n\theta$, so $s = 2\cos\frac{2\pi}{n}$ is a root of the monic integer polynomial $D_n(x) - 2$: an algebraic integer.

Hint 3. For the degree (for $n > 2$): $\varepsilon = e^{2\pi i/n}$ has degree $\varphi(n)$ (irreducibility of Φ_n), and it satisfies $\varepsilon^2 - s\varepsilon + 1 = 0$ over $\mathbb{Q}(s)$ but is not real. Hence $[\mathbb{Q}(\varepsilon) : \mathbb{Q}(s)] = 2$ and $[\mathbb{Q}(s) : \mathbb{Q}] = \varphi(n)/2$.

Problem 4

Hint 1. $\arctan k = \arg(1 + ki)$. If the identity held, exponentiating would give $\prod_k (\frac{1+ki}{|1+ki|})^{m_k} = \frac{1+16i}{\sqrt{257}}$. Square it and clear denominators to get an identity between Gaussian integers (move negative exponents to the other side).

Hint 2. You get $257\prod(1+ki)^{2m_k} = (1+16i)^2\prod(1+k^2)^{m_k}$, up to moving factors across. Now $257 = 1 + 16^2$ is a prime, so $1 + 16i = \pi$ is a Gaussian prime of norm 257.

Hint 3. Compare the exponent of π on both sides (unique factorisation in $\mathbb{Z}[i]$). No $1 + ki$ or $1 + k^2$ with $k \leqslant 15$ is divisible by π, because their norms $1 + k^2 \leqslant 226$ are not divisible by 257; and

$\bar\pi$ is not associate to π. The left side has π-exponent 1, the right side 2. Answer: no such integers exist.

Problem 5

Hint 1. $|1-\varepsilon^k| = 2\sin\frac{\pi k}{n}$, so you need $\sum_{k=0}^{n-1}\sin\frac{\pi k}{n} = \operatorname{Im}\sum_{k=0}^{n-1} e^{i\pi k/n}$.

Hint 2. The geometric sum equals $\frac{1-e^{i\pi}}{1-e^{i\pi/n}} = \frac{2}{1-e^{i\pi/n}}$, and $1-e^{i\alpha} = -2i\sin\frac{\alpha}{2}\,e^{i\alpha/2}$; its imaginary part is $\cot\frac{\pi}{2n}$.

Hint 3. Interpretation: $n|1-\varepsilon^k|$ is the perimeter of the star polygon $\{n/k\}$ obtained by joining every k-th vertex of the regular n-gon. So the sum is $\frac{1}{n}$ times the total perimeter of all star polygons $\{n/k\}$, $1 \leqslant k \leqslant n-1$ (twice the total length of all sides and diagonals, divided by n).

Problem 6

Hint 1. On $|z|=1$ we have $\bar z = 1/z$, so $1 = p(z)\overline{p(z)} = p(z)\,\bar p(1/z)$, where $\bar p$ has the conjugated coefficients. Multiply by z^n, $n = \deg p$, to get rid of negative powers.

Hint 2. With $p^*(z) = z^n\bar p(1/z)$ (a polynomial) the identity $p(z)p^*(z) = z^n$ holds on the unit circle, hence everywhere. So p divides z^n, i.e. $p = cz^m$, and $|c|=1$.

Hint 3. A trigonometric polynomial is $f(\theta) = e^{-iN\theta}p(e^{i\theta})$ for a polynomial p; apply the first part to p.

Problem 7

Hint 1. Pair k with $n-k$: for $|w|=1$, $w\neq 1$, one has $\frac{1}{1-w}+\frac{1}{1-\bar w} = 1$. Hence $\sum_{k=1}^{n-1}\frac{1}{1-\varepsilon^k} = \frac{n-1}{2}$.

Hint 2. $\frac{\varepsilon^k}{1-\varepsilon^k} = \frac{1}{1-\varepsilon^k} - 1$, so the first sum is $-\frac{n-1}{2}$.

Hint 3. $\frac{1}{1-e^{i\alpha}} = \frac12 + \frac{i}{2}\cot\frac{\alpha}{2}$. Taking imaginary parts in the second sum gives $\sum_{k=1}^{n-1}\cot\frac{k\pi}{n} = 0$ (also clear from the symmetry $k\leftrightarrow n-k$).

Problem 8

Hint 1. Start with the product formula: every root of z^n-1 is a primitive d-th root of unity for exactly one $d\mid n$, and the primitive d-th roots are $e^{2\pi ik/d}$ with $(k,d)=1$. Grouping the factors of $z^n-1 = \prod_k(z-\varepsilon^k)$ by order gives $\prod_{d\mid n}\Phi_d = z^n-1$.

Hint 2. Integrality by strong induction on n: $\Phi_n = (z^n-1)/\prod_{d\mid n, d<n}\Phi_d$, and division by a monic integer polynomial keeps integer coefficients.

Hint 3. $\Phi_p = 1+z+\cdots+z^{p-1}$; $\Phi_{p^2} = (z^{p^2}-1)/(z^p-1) = \Phi_p(z^p)$; for odd p, $\Phi_{2p}(z) = \Phi_p(-z)$ (the map $\zeta\mapsto-\zeta$ swaps primitive p-th and $2p$-th roots), while $\Phi_4 = z^2+1$.

Problem 9

Hint 1. Split the sum into $k\geqslant 0$ and $k<0$: it equals $\operatorname{Re}\Big(1+2\sum_{k\geqslant1}(re^{i\theta})^k\Big) = \operatorname{Re}\frac{1+re^{i\theta}}{1-re^{i\theta}}$. Multiply numerator and denominator by $1-re^{-i\theta}$.

Hint 2. For the series: $\sum_{k\geqslant1}\frac{r^k\cos k\theta}{k} = \operatorname{Re}\sum_{k\geqslant1}\frac{(re^{i\theta})^k}{k} = -\operatorname{Re}\log(1-re^{i\theta}) = -\frac12\ln(1-2r\cos\theta+r^2)$ (equivalently, integrate $\frac{P_r(\theta)-1}{2r}$ in r).

Hint 3. Let $r\to1^-$: the series $\sum\frac{\cos k\theta}{k}$ converges for $\theta\in(0,2\pi)$ (Dirichlet's test), so Abel's theorem applies, and $-\frac12\ln(2-2\cos\theta) = -\ln\big(2\sin\frac{\theta}{2}\big)$.

Problem 10

Hint 1. Expand $|z-\varepsilon^k|^2 = |z|^2 + 1 - 2\,\mathrm{Re}(\bar{z}\varepsilon^k)$ and use $\sum_k \varepsilon^k = 0$.

Hint 2. $\prod_k |z-\varepsilon^k| = |z^n - 1| \leqslant |z|^n + 1 = 2$ on $|z| = 1$, with equality iff $z^n = -1$. The maximum is 2.